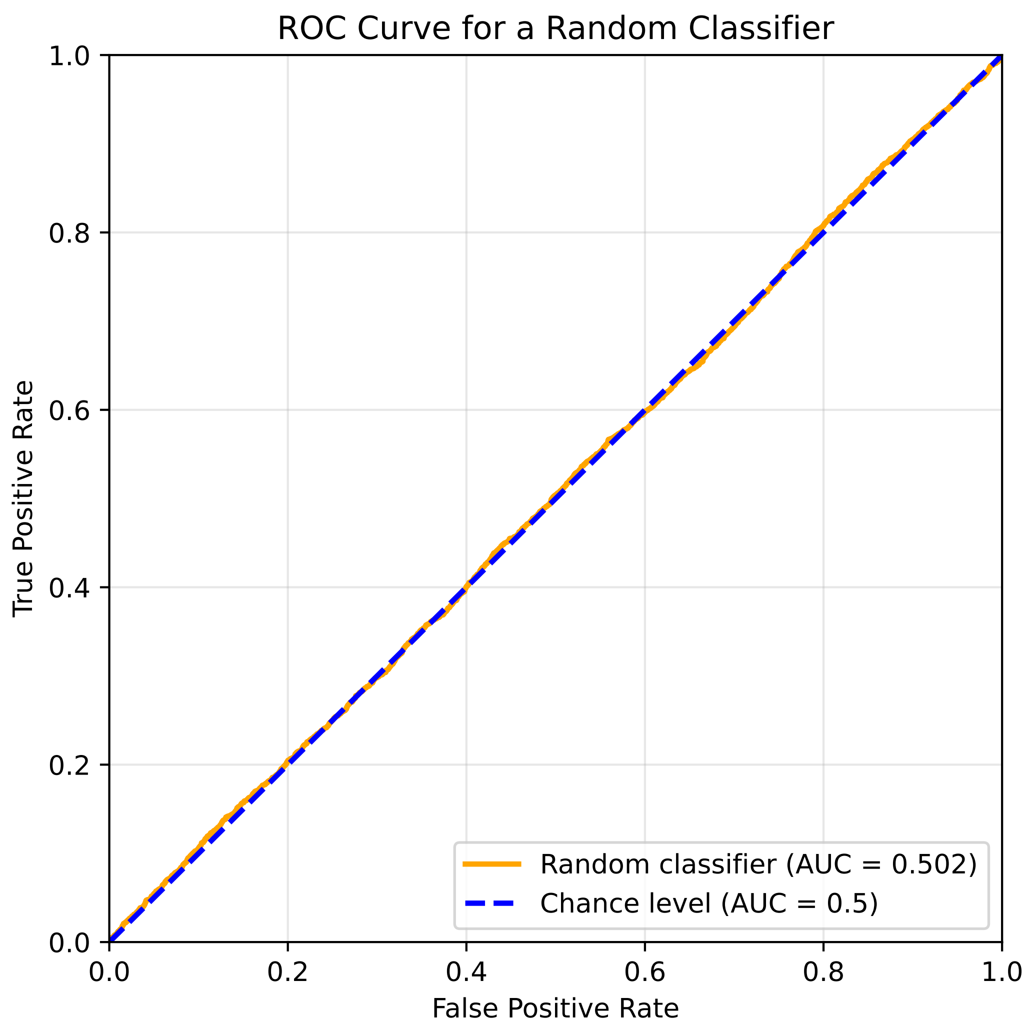
```python
import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics import roc_curve, auc

np.random.seed(1)

# Number of samples
n_samples = 10000
n_pos = 5000
n_neg = 5000

# Create true labels: 0 = negative, 1 = positive
y_true = np.array([1] * n_pos + [0] * n_neg)

# Shuffle the labels to mix positives and negatives
indices = np.random.permutation(n_samples)
y_true = y_true[indices]

# Generate random predictions (uniform between 0 and 1)
y_pred = np.random.rand(n_samples)

# Compute ROC curve and AUC
fpr, tpr, thresholds = roc_curve(y_true, y_pred)
roc_auc = auc(fpr, tpr)

# Plot ROC curve
plt.figure(figsize=(6, 6))
plt.plot(fpr, tpr, color='orange', lw=2, label=f'Random classifier (AUC = {roc_auc:.3f})')
plt.plot([0,1], [0,1], color='blue', lw=2, linestyle='--', label='Chance level (AUC = 0.5)')
plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.0])
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve for a Random Classifier')
plt.legend(loc="lower right")
plt.grid(alpha=0.3)
plt.show()

print(f"Area under ROC curve (AUC): {roc_auc:.4f}")
```




```
Area under ROC curve (AUC): 0.5019
```